# Do football betting markets know more than a model can learn? Auditing the closing line for calibration and information

Team **Team 32** (file: *Team-32-Proposal.pdf*) Date 2026-10-30

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## 1. Problem and research questions

Bookmakers publish closing decimal prices for home win, draw and away win, and the de-vigged probabilities implied by those closing prices are among the most accurate publicly available three-class forecasts in sport. They are almost always used as a rival to be beaten, or as a diagnostic of the bettor, rather than as an object to be audited. We invert the framing. The market is not our opponent but our most informative measuring instrument, and the questions worth asking are the ones it answers badly.

**RQ1.** After de-vigging, are market probabilities calibrated — does the reliability curve lie on the diagonal, and is the calibration slope equal to one? **RQ2.** Can a transparent model fitted only to pre-match information (no market columns in any feature set) match de-vigged closing log loss out of sample; if not, precisely where does the gap concentrate, and how much of it survives recalibrating the market itself?

This is not a football prediction project. In risk management, insurance pricing, credit scoring and clinical decision support, decisions are driven by predicted probabilities rather than predicted labels, and a model that classifies well but is badly calibrated is dangerous in each of those settings. Football gives us what those domains cannot: a high-frequency three-class outcome that is repeatedly observed, and a strong external forecaster with a known information set. A negative result here is informative only if the shortfall is localised and decomposable — which is exactly what we estimate.

## 2. Hypotheses

Each hypothesis is stated with the observation that would falsify it, fixed before any model is fitted.

**H1 — The market is under-dispersed, not unbiased.** De-vigged closing probabilities have a calibration intercept near zero but a *slope greater than one*: the reliability curve lies above the diagonal at low predicted probabilities and below it at high, so longshots are over-priced and favourites under-priced. This is the favourite–longshot bias (Ali 1977; Cain, Law & Peel 2000) and we do not claim it is new. The open question is whether it *survives de-vigging*: Štrumbelj (2014) shows that some margin-removal methods remove it while proportional normalisation does not. We therefore report the slope and the tail-bin bias under proportional, Shin and power de-vigging, and we pre-register the prediction that the bias is largest under proportional de-vigging. *Falsified if the proportional-de-vigged slope is at most 1.*

**H2 — A transparent model does not close the gap, and the residual is not negligible.** Two one-sided claims, both pre-specified. (a) M3 (Dixon–Coles) does not beat the market: *falsified if the 95% cluster-bootstrap interval on the paired per-match log-loss difference excludes zero in the model's favour.* (b) The residual gap is practically non-zero: *falsified if the 90% interval lies entirely within ±0.01 nats per match* (about 1% relative), our pre-specified equivalence margin. If an interval is wider than the margin we report the estimate descriptively and claim neither superiority nor equivalence.

**H3 — The gap is localised, not diffuse.** The per-match gap (market minus model; positive means the model is worse) is larger for matches involving a newly promoted side, and for the first six matchdays of a season; both indicators are estimated jointly with league-season fixed effects. The mechanism is that each signals stale strength information — the market observes transfers, preselection news and spending, whereas our model observes only earlier scorelines. *Falsified if either coefficient is at most 0.* We report a matchday-window sweep (1–3, 1–6, 1–10) so that "six" is a reported robustness curve rather than an arbitrary threshold.

**H4 — The gap is information, not miscalibration.** Replacing the market benchmark MK with a recalibrated market MK* — a two-parameter temperature-and-bias map fitted in-sample and applied out-of-sample — changes the estimated model–market gap by *less than 25%. Falsified if recalibration closes more than a quarter of the gap*, which would mean we are losing to a fixable defect in the price rather than to superior information. This question is the reason to run RQ1 at all: it decides how the rest of the results should be read.

## 3. Data

- **Source.** football-data.co.uk — publicly downloadable, no registration and no API key. One CSV per league-season, match-level, cross-checked against the archive's own field documentation.
- **Window.** Six leagues (English Premier League, La Liga, Serie A, Bundesliga, Ligue 1, Eredivisie), seasons 2016/17–2024/25: **18,865 matches**. The 2016/17 season (2,132 matches) serves as the initial training block, leaving **16,733 evaluated matches across 48 (league, season) blocks**.
- **Benchmark.** We use Pinnacle's *closing* price (columns PSCH/PSD/PSCA). The archive distinguishes pre-closing prices, collected on Friday afternoons, from closing prices, which carry a C suffix; a pre-closing benchmark would understate the market and mechanically shrink the very gap we are trying to measure, since pre-closing margins are larger and rise over time (median Bet365 overround in the Premier League: 2.59% in 2016/17, 5.51% in 2024/25). Pinnacle closing is present for at least 99.0% of matches in every league-season of the window, and carries the lowest margin of any single bookmaker in the archive (median overround 2.05% in the Premier League in 2016/17, 2.94% in 2024/25). Market-maximum closing prices (MaxCH family, available from 2019/20, median overround 0.23% in the Premier League) are used as an almost margin-free robustness benchmark. The 2025/26 season is excluded because Pinnacle closing coverage falls to roughly 55%.
- **Target.** Three-class outcome: home win, draw, away win.
- **Known challenges.** (i) Strong chronological dependence, so any random split would leak the future; (ii) promotion and relegation make the set of team identities non-stationary; (iii) two curtailed seasons inside the window — Ligue 1 2019/20 ends at 279 matches and Eredivisie 2019/20 at 232; (iv) football is intrinsically noisy, top-1 accuracy is bounded near 50%, so accuracy is an inadequate metric and strictly proper scoring rules must be the evaluation currency; (v) team names and league codes change spelling across seasons and must be harmonised before any pooling.

## 4. Models

The ladder is built so that adjacent rungs differ in exactly one factor, which makes each step a single interpretable comparison. **No market column enters any model arm**, by construction: market columns are used only to construct the benchmarks MK and MK*.

- **M0 — Prior baseline.** Expanding league-level outcome frequencies estimated from strictly earlier matches: no covariates, no current-season information.
- **M1 — Elo ratings** mapped to three-way probabilities by an ordered logit (home/draw/away has a natural ordering, and proportional odds yields a single interpretable rating effect).
- **M2 — Multinomial logit** on time-decayed goal-based features: goal difference, home and away splits, rest days, promoted flag.
- **M3 — Dixon–Coles** bivariate Poisson with exponential time-decay $\xi$, home advantage $\gamma$ and low-score dependence $\rho$; the primary structural model, and the arm on which H2 is tested.
- **M4 — Gradient boosting** on the same pre-match feature set as M2. This is a *flexibility probe*, not a bid for accuracy: if M4 and M3 share the same residual gap, that gap is a deficiency of information rather than of model class, which is what makes the H3 mechanism claim interpretable.
- **MK — Market benchmark.** De-vigged Pinnacle closing probabilities, with proportional normalisation primary and Shin's and the power method as robustness.
- **MK* — Recalibrated market.** MK passed through a two-parameter logit recalibration fitted in-sample and evaluated out-of-sample; the instrument for H4.

## 5. Evaluation

**Protocol.** Expanding-window rolling origin by season: for target season $s$ and league $\ell$, every model is fitted on matches in league $\ell$ strictly before $s$ and used to predict every match of $s$. That gives 8 target seasons × 6 leagues = 48 blocks and 16,733 matches, with each block evaluated exactly once. Hyperparameters ($\xi$, $\rho$ and the boosting settings) are selected by an inner expanding-window validation that sees only pre-$s$ data, so no outer block is ever used for tuning.

**Inference.** All claims are paired per-match differences on the same matches, so the interval is on the difference itself rather than on two marginal scores. Intervals are cluster-bootstrap over the 48 (league, season) blocks (10,000 resamples), with cluster-robust standard errors as a cross-check. Because 48 clusters is a modest number and matches within a block share a squad pool and a set of estimated strengths, we report the estimated design effect alongside every interval: at this cluster count the detectable difference is governed by within-block correlation, not by the number of matches. Per-league and per-season breakdowns are declared descriptive in advance, so that no result is promoted after the fact on the strength of a favourable subset.

**Metrics.** Log loss primary; Brier score and ranked probability score secondary. Calibration is assessed by reliability diagrams, by a calibration slope and intercept estimated per outcome class through one-versus-rest logistic recalibration (a slope above 1 indicates an under-dispersed forecast), by expected calibration error, and by a bias-versus-predicted-probability plot that carries H1's regional claim. All are strictly proper scoring rules (Gneiting & Raftery 2007), so calibration cannot be improved merely by sharpening the output.

## 6. Expected contribution

- A calibration audit of real closing prices that reports its own **dependence on the de-vigging method** instead of silently choosing one.
- A **localised** account of where a transparent model falls short — promoted sides and early-season fixtures — rather than a single aggregate accuracy number.
- A **decomposition of the market's advantage** into a calibration component and an information component, which is the one comparison the existing literature on either side does not make.
- A reusable match-level dataset and a reproducible pipeline (pinned versions, checksummed downloads, fixed seeds) delivered as the appendix source code.

Negative results are the designed outcome here. Whether or not a model beats the market, the project still delivers a calibration map of the price and a diagnosis of where its information lives. This project is newly initiated for DSC5001C63 and does not reuse data, code or analysis from any other course.

## 7. Team and work packages

| Work package | Content | Owner |
|---|---|---|
| **Data** | Six-league ingestion, name harmonisation, checksummed snapshot, coverage matrix | ZHOU Yixuan |
| **M1–M2** | Elo ratings and multinomial logit | WANG Xingshuai |
| **M3–M4** | Dixon–Coles with ξ/γ/ρ estimation; boosting probe | SHU Peixuan |
| **MK / MK*** | De-vigging under three methods; calibration diagnostics; bootstrap intervals | SUN Peng |
| **Outputs** | Literature benchmark, slides, all-member recording, appendix | WAN Yihui |

## 8. Timeline

| Period | Milestone |
|---|---|
| **30 Oct – 1 Nov** | Proposal submission; six-league ingestion and the (league × season) coverage matrix. |
| **2 – 9 Nov** | Rolling-origin harness with the inner validation loop; M0–M2; first end-to-end pass. |
| **10 – 16 Nov** | M3 numerical stabilisation (ξ, γ, ρ); M4; MK under three de-vigging methods and MK*. |
| **17 – 23 Nov** | H1–H4 analysis; calibration diagnostics; cluster bootstrap; core figures. Slides skeleton drafted in parallel. |
| **24 – 27 Nov** | Slides (10 pages), all-member recording (target 26–27 Nov, backup 28 Nov), appendix code and reproducibility manifest. |
| **29 Nov** | Report submission. |

## 9. References


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